STUDENT REFERENCE & INTERNAL USE NOTICE

This protocol documents standard in-vitro Emulsification Index (E24) assay procedures for biosurfactant- and exopolysaccharide-producing microorganisms used in HOPE Lab research and is PROVIDED ONLY FOR THE HOPE LAB MEMBERS for training and internal reference. External users should independently assess method suitability, verify current standards and chemical safety for reagents, and obtain required biosafety authorization and other essential approvals. This document does not replace an institutionally approved SOP or method-specific risk assessment.

TITLE

Emulsification Index (E24) Assay for Biosurfactant- and Exopolysaccharide-Producing Microorganisms

PRINCIPLE

The emulsification index after 24 hours (E24) is an operational measure of the persistence and stability of an oil–water emulsion under defined assay conditions. A test material, typically a cell-free culture supernatant or partially purified fraction, is combined with a selected hydrophobic phase (oil) and subjected to standardized high-energy vortex mixing. Following an undisturbed 24-hour settling period, the vertical height of the persistent emulsion layer is measured and expressed as a percentage relative to the total liquid-column height.

E24 values depend strongly on oil composition, test-material concentration, medium matrix, tube geometry, mixing energy, ambient temperature, and interface-reading conventions. Both low-molecular-weight biosurfactants and high-molecular-mass biopolymers such as exopolysaccharides (EPS) can stabilize emulsions. Consequently, an elevated E24 value is a comparative phenotypic indicator and does not independently identify the responsible molecule, quantify biosurfactant concentration, or prove interfacial surface-tension reduction.

Learning Outcomes

- Explain what the E24 index measures operationally and its analytical boundaries.
- Identify the necessity of a matched uninoculated-medium matrix control and a positive surfactant reference.
- Calculate E24 (%) accurately using a clearly defined and consistent liquid-height measurement convention.
- Control and record critical physical variables (oil identity, tube diameter, mixing duration, and temperature).
- Report biological versus technical replicate variation and potential sample clarification effects.

OBJECTIVE & SCOPE

Objective: To establish standardized principles, mathematical calculations, experimental controls, and reporting conventions for estimating the 24-hour emulsification index (E24) of microbial culture-derived materials.

Scope: Applicable to cell-free culture supernatants, exopolysaccharide solutions, and characterized fractions evaluated under consistent oil, volume-ratio, vessel-geometry, and mixing conditions. Whole-cell suspensions are excluded unless the study specifically investigates cell-bound emulsification using a separately validated protocol.

CORE CONCEPTS & CALCULATIONS

- **E24 Calculation Formula:** Let h_e represent the vertical height (mm) of the stable emulsion layer after 24 hours, and h_t represent the total height (mm) of the complete liquid column (aqueous + emulsion + residual oil phases), measured with the exact same boundary convention:
  $E24\ (\%) = (h_e / h_t) \times 100$
  Transient surface foam above the upper oil meniscus must be excluded from both h_e and h_t.
- **Choice & Standardization of Oil:** Different hydrocarbons and vegetable oils (e.g., kerosene, n-hexadecane, olive oil, soybean oil) yield markedly different E24 values for the identical sample. Use one defined oil lot and grade across a comparative study; never compare E24 values generated with different oils directly.
- **Matched Matrix Control Rule:** Uninoculated culture medium processed identically to test cultures assesses background emulsification from medium peptides or salts. Report matrix-control E24 separately alongside sample E24. Do not routinely subtract matrix-control E24 from sample E24, as E24 is a non-linear ratio.
- **Positive Reference & Replicates:** Include a documented positive control (e.g., 0.1–1.0% w/v Sodium Dodecyl Sulfate [SDS] or Tween-80) to confirm assay responsiveness. Distinguish independent biological preparations from repeated technical measurements.

MATERIALS & EQUIPMENT

Reagents & Consumables

- Cell-free culture supernatant or purified EPS/biosurfactant fraction
- Matched uninoculated culture medium (Matrix Control)
- Defined hydrophobic phase / oil (e.g., kerosene, n-hexadecane, or food-grade vegetable oil of documented grade)
- Positive reference surfactant (e.g., SDS or Tween-80 standard solution)
- Matched flat-bottom or uniform cylindrical glass test tubes (identical internal diameter)

Equipment & Instrumentation

- High-speed Vortex Mixer (capable of reproducible maximum speed, e.g., 2,500–3,000 rpm)
- Refrigerated Centrifuge (for cell removal, e.g., 8,000–10,000 × g at 4°C)
- Precision Digital Vernier Calipers or Millimeter Scale
- Temperature-monitored Workstation / Incubator (e.g., 25°C or 30°C)
- Fixed-position Digital Camera Setup & Tube Rack

STEP-BY-STEP PROCEDURE

Step 1: Test-Material Clarification & Matrix Control Preparation

Harvest stationary-phase microbial cultures and remove cells by centrifugation (e.g., 8,000–10,000 × g for 15 min at 4°C). Carefully collect the cell-free supernatant. Note: If membrane filtration (0.22 µm) is used, verify and document that high-molecular-mass EPS or aggregated biosurfactants are not retained on the filter. Process an uninoculated medium aliquot identically as the matched matrix control.

Step 2: Dispensing Aqueous & Oil Phases

Select clean, grease-free, matched cylindrical glass tubes of identical internal diameter. Pipette a defined volume of cell-free test sample (e.g., 2.0 mL) into each tube, followed by an equal volume (2.0 mL, 1:1 v/v ratio) of the selected oil. Prepare matched matrix-control tubes and positive-control surfactant tubes in parallel.

Step 3: High-Energy Vortex Mixing

Cap each tube securely. Vortex each tube individually at maximum validated speed (e.g., 2,500–3,000 rpm) for exactly 2 minutes without interruption to ensure uniform shear energy and droplet dispersion across all samples and controls.

Step 4: Undisturbed Settling Period

Immediately place the vortexed tubes upright in a stable rack at a constant, monitored ambient temperature (e.g., 25 ± 2°C). Leave the tubes completely undisturbed for exactly 24 hours.

Step 5: Layer Boundary Identification & Measurement

At 24 hours, inspect tubes against a high-contrast background without tilting or agitating. Measure the vertical height of the persistent, opaque emulsion layer (h_e) and the total liquid-column height (h_t, from tube bottom meniscus to top oil meniscus, excluding transient foam) in millimeters using digital calipers. Capture standardized photographs for archival verification.

Step 6: Calculation & Replicate Reporting

Compute E24 (%) = (h_e / h_t) × 100 for each replicate. Report the mean E24 (%) and standard deviation (SD) across at least three independent biological replicates, alongside the separate E24 values of the uninoculated matrix control and positive reference.

CRITICAL FACTORS AFFECTING RESULTS

Variable	Effect of Inadequate Control	Required Control Standard
Oil Identity & Grade	Emulsion stability varies drastically across hydrocarbon/lipid types	Use one defined oil lot/grade within a study; report identity
Test-Material Matrix	Medium proteins/salts create background emulsification	Include matched uninoculated matrix control; report separately
Mixing Energy & Time	Inconsistent shear alters droplet size and layer persistence	Vortex at fixed maximum rpm for exactly 2.0 min per tube
Vessel Geometry	Varying internal tube diameters distort relative layer heights	Use matched tubes of identical internal diameter & volume
Temperature & Timing	Coalescence and phase separation depend on temp and time	Incubate undisturbed at monitored temp; read at exact 24 h
Sample Clarification	0.22 µm filtration can adsorb high-molecular-mass EPS/micelles	Prefer centrifugation or validate filtration recovery first
Boundary Definition	Ambiguous interfaces or foam bias height measurements	Apply strict meniscus/foam exclusion rules; retain photos

QUALITY CONTROL & TROUBLESHOOTING

Quality Control (QC): Include a matched uninoculated medium control and a positive surfactant reference (e.g., SDS) in every batch. Record raw layer heights (h_e and h_t in mm), oil identity, temperature, clarification method, and high-resolution photographs.

Observation	Possible Explanation	Corrective Action
No persistent emulsion layer	Sample lacks emulsifying activity under chosen oil/conditions	Confirm positive control formed emulsion; report 0% / low E24
Matrix control shows emulsion	Medium components (e.g., peptones, yeast extract) stabilize oil	Report control E24 separately; do not apply simple subtraction
High replicate variation	Uneven vortex mixing, dirty glassware, or tube diameter mismatch	Degrease tubes thoroughly; standardise vortex timing & speed
Unclear emulsion boundary	Turbid aqueous phase, gradual gradient, or persistent foam	Follow predefined meniscus convention; document with photos
Activity drops after filtration	High-molecular-mass EPS or aggregates retained on filter	Use cell-free supernatant from centrifugation; note filter effect

BIOSAFETY & ADDITIONAL CONSIDERATIONS

- **Biosafety & Chemical Waste:** Handle microbial cultures and culture supernatants according to institutional biosafety containment (BSL-1/BSL-2 as assessed). Manage hydrocarbons, solvents, and SDS according to current Safety Data Sheets (SDS); collect oil–water emulsions in designated chemical/biohazard waste containers rather than sink disposal.
- **Scientific Reporting Scope:** Report E24 strictly as a comparative, condition-dependent index of emulsion stability. Do not report E24 alone as proof of biosurfactant chemical identity, molar concentration, critical micelle concentration (CMC), or surface-tension lowering.

REFERENCES & RELATED DOCUMENTS

Ali F, Das S, Hossain TJ, Chowdhury SI, Akter ZS, Das T, et al. (2021). Production optimization, stability, and oil emulsifying potential of biosurfactants from selected bacteria isolated from oil contaminated sites. Royal Society Open Science, 8, 210301. doi:10.1098/rsos.210301

Zobaer M, Ali F, Anwar MN, Bappi MSH, Bakar TB, Hossain TJ, Khan MS, Zawad ANMS. (2024). Isolation of biosurfactant producing bacteria from oil-spilled soil and characterization of their secreted biosurfactants in pathogen-inhibition and oil-emulsification. International Journal of Pharmaceutical Sciences and Research.

Tarannum N, Hossain TJ, Ali F, Das T, Dhar K, Nafiz IH. (2023). Antioxidant, antimicrobial and emulsification properties of exopolysaccharides from lactic acid bacteria of bovine milk: insights from biochemical and genomic analysis. LWT - Food Science and Technology, 186, 115263.